

AEQUITAS NODE OPERATOR GUIDE

Version 1.0 · June 2026 · aequitas.digital

Complete step-by-step guide · No prior blockchain experience required · Estimated time: 20–30 min

What is an Aequitas Node?

An Aequitas node is a program that runs in the cloud and participates in the Aequitas network. It keeps a copy of the entire blockchain, validates who is a registered human, and produces new blocks. The more nodes exist, the more decentralized and resilient the network becomes. As a reward for running a node, you receive a daily share of all protocol fees — **automatically at 20:00 Berlin time (CEST/CET)**, no further action required.

Before You Start — What You Need

- Aequitas account:** Register as a human via the Android app. You need a wallet address to receive rewards.
- GitHub account (free):** Go to github.com — you need this to fork the Aequitas code.
- Railway account (free):** Go to railway.app and sign in with GitHub. No server or command line required.
- Dedicated node wallet:** A separate MetaMask wallet for your node (NOT your personal AEQ wallet). Export its private key: MetaMask → Account Details → Show Private Key → enter password → copy. **Keep this key strictly private.**
- 10–30 minutes of your time.** Railway does most of the work automatically.

Environment Variables — Complete Reference

Collect these values before deploying. Enter them in Railway → Variables.

SECURITY: Your RELAYER_PRIVATE_KEY is like a master password. Never share it.

Variable	Required?	What to set
DATABASE_URL	YES	Auto-injected by Railway when PostgreSQL is in the same project. Format: postgres://user:pass@host:5432/dbname
RELAYER_PRIVATE_KEY	YES	Private key of your dedicated node wallet (starts with 0x, 66 chars). MetaMask → Account Details → Show Private Key
RELAYER_ADDRESS	Recommended	Public address matching RELAYER_PRIVATE_KEY (0x, 42 chars). Prevents startup errors.
NODE_OPERATOR_WALLET	For rewards	Your Aequitas HUMAN wallet address (the one you registered with). This receives daily validator rewards at 20:00 Berlin time.
PEER_SECRET	For multi-node	Shared secret string — ALL nodes in the network must use the SAME value. Get this from the network operator. Do not share publicly.
SELF_URL	For multi-node	Your node public URL: https://YOUR-NAME.up.railway.app Find in Railway → Settings → Networking → Public Networking.
PRIMARY_NODE_URL	For multi-node	Set to: https://aequitas.digital Your node registers here automatically on startup.
NODE_KEY	Optional	Base64-encoded libp2p private key for stable peer identity. If not set: auto-generated on first start, printed to stderr as "SAVE THIS AS NODE_KEY ENVIRONMENT VAR: " Copy that value here to keep a stable node ID across restarts.

Variable	Required?	What to set
IS_PRIMARY_NODE	NO	Leave unset (or false). Only the official primary node uses true. Setting true on secondary nodes causes double pool distributions.
RESET_STATE	NO	DANGEROUS: wipes the entire database on restart. Development only.

Step-by-Step Deployment on Railway

Step 1 — Fork the Repository

Open github.com/hanoi96international-gif/Aequitas in your browser. Click **Fork** in the top-right → **Create fork**. GitHub creates a copy under your account.

Step 2 — Create PostgreSQL Database

Go to railway.app and sign in with GitHub. Click **New Project** → **+ New** → **Database** → **Add PostgreSQL**. Railway creates the database and will auto-inject `DATABASE_URL` when you add your node.

Step 3 — Deploy Your Node

In the same Railway project: **+ New** → **GitHub Repo** → select your Aequitas fork. Railway detects the Dockerfile automatically. Click **Deploy Now**.

Step 4 — Set Environment Variables

Click your Aequitas service → **Variables**. Add at minimum:
`RELAYER_PRIVATE_KEY = 0xYOUR_PRIVATE_KEY`
`RELAYER_ADDRESS = 0xYOUR_NODE_WALLET_ADDRESS`
`NODE_OPERATOR_WALLET = 0xYOUR_HUMAN_WALLET`
`PEER_SECRET = get-from-network-operator`
`SELF_URL = https://YOUR-RAILWAY-DOMAIN.up.railway.app`
`PRIMARY_NODE_URL = https://aequitas.digital`
Save → Railway auto-redeploys.

Step 5 — Get Your Public URL

Railway → Settings → Networking → **Generate Domain**. Open `https://YOUR-URL/api/status` — you should see JSON with `height` climbing every ~6 seconds.

Step 6 — Verify Success

Check deploy logs for:
✓ API Server listening on port 8080
[SYNC] Connected to peer `https://aequitas.digital`
[NODE] Registered node operator wallet: 0x...
Block height should match the primary node within 1-2 blocks.

Step 7 — Earn Rewards

Rewards are distributed automatically every day at 20:00 Berlin time (CEST/CET). The Validators Pool (40% of all protocol fees) is split proportionally among all registered node operators. Keep your node running — no further action needed.

Optional: Register Validator Key (Decentralized Auth)

Instead of a shared PEER_SECRET, you can register your node with a cryptographic signature. This proves you control the node private key without sharing a secret.

1. Request a challenge:

```
GET https://aequitas.digital/api/peers/challenge?address=0xYOUR_RELAYER_ADDRESS
```

2. Sign the returned challenge string with your node private key (RELAYER_PRIVATE_KEY) using MetaMask or ethers.js personal_sign.
3. POST to https://aequitas.digital/api/peers/register with your signing_address and signature. The challenge is valid for 90 seconds.

On the website Network → Run a Node tab, use the "Sign with MetaMask & Register" button for the easiest flow.

Troubleshooting

Symptom	Likely Cause	Solution
Block height stays at 0	PRIMARY_NODE_URL not set	Set PRIMARY_NODE_URL=https://aequitas.digital and also set SELF_URL to your own node URL. Redeploy.
DATABASE_URL error	Wrong connection string	Format: postgres://user:pass@host:5432/dbname — ensure PostgreSQL is running.
"no code at address" in logs	V7 contract not deployed yet	Normal on first start — node auto-deploys V7. Wait a few seconds.
No validator rewards	NODE_OPERATOR_WALLET not set	Add NODE_OPERATOR_WALLET=0xYOUR_HUMAN_WALLET to Railway Variables.
Railway shows "Application error"	Build or startup failure	Check Deploy Logs. Most common: missing DATABASE_URL or wrong RELAYER_PRIVATE_KEY format (must start with 0x).
Port 8080 not reachable (Docker)	Firewall config	Open TCP port 8080 inbound in your firewall or cloud security group.

Questions & Feedback

Open an issue on GitHub (github.com/hanoi96international-gif/Aequitas) or visit aequitas.digital for the latest information. BETA feedback on node setup, performance, and documentation gaps is especially welcome.

Aequitas Chain · Chain ID 1926 · aequitas.digital · UBI distributions: every day at 20:00 Berlin time (CEST/CET)